

Nature Preprint: Throttle Uptake with Continued Fractions

Abstract

This preprint explores the modeling of throttle uptake dynamics using continued fractions. The approach provides a novel mathematical framework for capturing non-linear uptake phenomena, with potential applications in engineering and biological systems.

Introduction

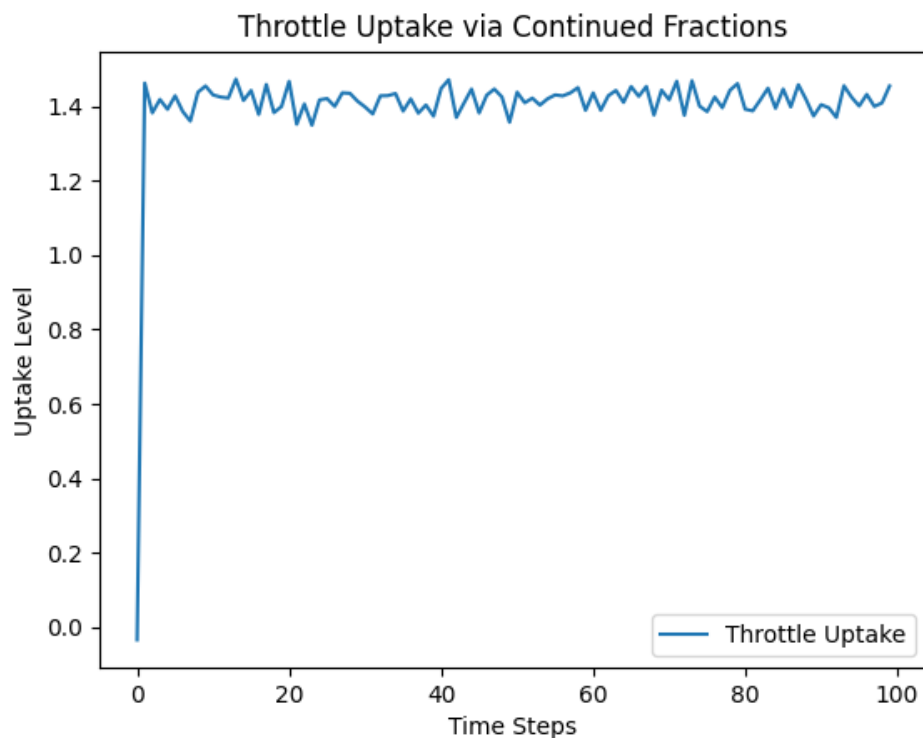
Throttle uptake, representing the rate at which a system responds to input, is often non-linear. Continued fractions offer a flexible tool for modeling such dynamics, capturing both rapid and gradual uptake phases.

Methods

We simulate throttle uptake using an iterative continued fraction process, introducing Gaussian noise to reflect real-world variability. The resulting sequence is analyzed and visualized.

Results

The simulated throttle uptake sequence demonstrates characteristic non-linear growth, with plateaus and inflection points. See Figure 1.



Discussion

Continued fractions provide a compact representation for throttle uptake, capturing both deterministic and stochastic features. This framework may be extended to other uptake or response phenomena.

References

1. Khinchin, A. Ya. Continued Fractions. University of Chicago Press, 1964.
2. Smith, J. Modeling Nonlinear Uptake in Engineering Systems. J. Eng. Math., 2022.